Athlete's foot or Tinea pedis (Dermatophytoses)	*T. mentagrophytes, Trichophyton rubrum*	Direct contact	redness or blisters, peeling or cracking skin	topical antifungal ointments, itraconazole, terbinafine
Tinea capitis or scalp ringworm (Dermatophytoses)	*Trichophyton tonsurans, T. schoenleinii, T. violaceum*	Direct contact	round patches of dry scale, alopecia	Griseofulvin, Terbinafine
Mucormycosis	*Mucor and Rhizopus*	Air borne	Opportunistic pathogen	Amphotericin B

10.3 IMMUNITY

Immunity refers to the ability of the immune system to defend against diseases caused by microbes or foreign substances which are products like toxins from microbes. Immunity depends upon various factors like host resistance, dosage of organism injected and virulence of the organisms.

10.3.1. Classification

Immunity is mainly classified into innate and acquired immunity. Innate or nonspecific or natural immunity refers to the basic resistance to disease that an individual is born with. Acquired or specific or adaptive immunity requires the activity of a functional immune system, involving cells called lymphocytes and their products. Innate defense mechanisms provide the first line of defense against invading pathogens until an acquired immune response develops. In general, most of the microorganisms encountered by healthy individual are readily cleared within a few days by non-specific defense mechanisms without enlisting a specific immune response. When an invading microorganism eludes the nonspecific host defense mechanism, a specific immune response occurs.

10.3.1.1. Innate (Natural) Immunity

Innate immunity may be considered at the level of the species, race or individual. Species immunity refers to the total or relative refractoriness to a pathogen, shown by all members of a species. For instance, all human beings are totally insusceptible to plant pathogens and to many pathogens of animals such as rinderpest.. This immunity is something a person obtains by birth, for the reason that he belongs to the human species. The mechanisms of species immunity may be due to physiological and biochemical differences between the tissues of the different host species, which determine whether or not a pathogen can multiply in them.

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